Omar abushmais

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EDUCATION

Al-Hussein Technical University (HTU) — **Amman, Jordan**
Bachelor of Science in Computer Science | GPA: 3.69 — Sep 2026

EXPERIENCE

Safwa Islami Bank — **Amman, Jordan**
AI/Automation Developer Intern — Feb 2026 – Sep 2026

- Designed and built a Retrieval-Augmented Generation (RAG) chatbot that replaced manual searches through the bank's official releases, letting employees get accurate, source-grounded answers in seconds instead of digging through documents.
- Built an automated ingestion pipeline (n8n, PostgreSQL, Docker) that processes PDF, Excel, and DOCX files into vector embeddings automatically, eliminating manual document handling.
- Engineered a question-answering workflow combining vector search, semantic caching, and Cohere reranking to cut redundant model calls and speed up response times, with live web search as a fallback for out-of-scope queries.
- Refined prompts across intent validation, query construction, answer synthesis, and session-title generation to improve response accuracy and consistency.
- Developed a React-based chat interface with session tracking and source traceability, and added centralized logging/error handling to keep the system reliable in production.

Bitknowledgy — **Amman, Jordan**
Front-end developer intern — Oct 2025 – Dec 2025

- Built 10–15 responsive, production-ready front-end pages in an MVC system (HTML, Bootstrap, JavaScript, C#) for an enterprise-facing platform.
- Connected UI components to back-end controllers and models so dynamic tables and forms displayed live data accurately.
- Added Arabic/English localization support to improve accessibility and simplify future content updates.
- Fixed inherited UI, logic, and data-fetching bugs, reducing rework for the rest of the team.

PROJECTS

Real-Time Bidding System (Full-Stack Web Application)
React (Vite), Node.js, Express.js, PostgreSQL, REST APIs

- Developed a full-stack real-time bidding platform with React (Vite) frontend and Node.js/Express backend, enabling users to create auctions, place bids, and track live activity with dynamic, responsive UI and real-time updates.
- Designed a scalable RESTful backend with authentication, auction logic, bid validation, user management, role-based access control, and secure API routing.
- Created a normalized PostgreSQL database schema with relationships, constraints, and transactional logic to ensure data integrity and consistent bidding behavior.
- Implemented secure authentication workflows using password hashing, JWT-based sessions, protected routes, and role-based access control.
- Managed version control and collaborative development using Git and GitHub for commits, branching, and clean code history.

GitHub:
Client: https://github.com/Omarabushmais/bidrush-client
Server: https://github.com/Omarabushmais/bidrush-server

Patient Appointment System
HTML, CSS, JavaScript, Python (FastAPI), SQL Alchemy, MySQL

- Implemented a responsive, dynamic frontend using HTML/CSS and JavaScript, integrating all pages with backend endpoints to fetch, update, and display real appointment data.
- Built a complete backend using FastAPI and SQL Alchemy, implementing secure REST APIs for authentication, scheduling, appointment logic, password resets, and user management.
- Designed and developed a relational MySQL database with proper relationships, constraints, time-slot logic, and data validation to prevent double bookings and ensure consistency.
- Implemented secure authentication workflows using password hashing, token-based sessions, permission checks, and protected API routes to enforce role-based access control at the code level.

Co-Working Space Booking DBMS
C, Multithreading, Custom DBMS Logic, Mutex Locks, Hash Indexing

- Built a full custom mini-DBMS in C using CSV files for persistent storage, with in-memory dynamic tables supporting resizing, indexing, and efficient lookups.
- Implemented CRUD operations for users, workspaces, resources, bookings, and payments with validation, relationship checks, and constraint rules.
- Designed a hash-table indexing system for O(1) ID lookups, enabling fast access across all tables.
- Added concurrency control by integrating pthread mutex locks across all DB operations (user, workspace, resource, booking, payment layers) to ensure thread-safe execution.
- Developed multithreaded features including concurrent menu operations and a custom race condition demo showing safe vs unsafe access using shared counters and mutexes.

SKILLS

Languages & Web: C, C++, Python, Java, C#, SQL, HTML, CSS, JavaScript
Frameworks: React, Node.js, Express.js, FastAPI, ASP.NET Core MVC, Git, GitHub
AI & Automation: RAG, Prompt Engineering, PostgreSQL/pgvector, n8n, Docker
Game Development: Unity, Unreal Engine, Blender